Fifteen Mechanisms That Did Not Work on a Perpetual-Futures DEX

A Negative-Results Census, with Reproducible Microstructure Measurements

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August 2026

Abstract

Over six months and roughly ten thousand tested configurations on Hyperliquid, a fully on-chain perpetual-futures exchange, we attempted to monetise short-horizon predictive structure in the limit order book and adjacent mechanisms, and failed fifteen times. This paper reports the graveyard deliberately: negative results in trading research are rarely published, so each generation of researchers re-derives the same walls at full cost. We organise the record by *economic mechanism* (who is on the other side, and why they would lose) and by *death reason* (arithmetic, mechanism absent, underpowered, or blocked), a distinction that decides whether a question is closed or merely unresolved. The common signature of every refutation is stated and quantified: the predicted move is comparable to the round-trip cost. Alongside the graveyard we report descriptive measurements of the venue that are, to our knowledge, undocumented in the academic literature: the median pair quotes a half-spread 17–19× that of BTC; spread and depth are uncorrelated in cross-section ($\rho = -0.107$); the median pair carries under \$100 of notional at the touch and only 4 of 177 pairs hold \$5,000; funding settles hourly; and the venue's candle history is capped at $\sim$5000 bars per interval, making fine-grained history unrecoverable unless accumulated live. A frozen data slice and a one-command script reproducing the headline measurements ship with the paper. We close with the transferable result: a twenty-minute expected-move-to-cost calculation, run before any infrastructure is built, would have redirected the entire programme in week one.

Keywords: negative results, market microstructure, perpetual futures, decentralised exchanges, transaction costs, adverse selection, reproducibility

JEL Classification: C58, G12, G14, G23

Contents

*Working paper, preliminary draft — comments welcome. All estimates are from an internal empirical record consolidated 2026-08-12; section references of the form §n in tables refer to that record and are retained for provenance. Please do not cite specific numbers without checking for a newer version.

1 Introduction

The empirical starting point of this programme was a strong positive result, documented in a companion paper: order-book imbalance on BTC, ETH and SOL perpetuals predicts the direction of the mid-price with a rank information coefficient of ≈0.45 at horizons of one to five seconds, universally across symbols and volatility regimes. The result survives every statistical control we ran at it. What follows here is the record of attempting to turn that structure, and fourteen adjacent mechanisms, into realised profit — and failing every time.

Negative results of this kind are close to unpublishable in the standard venues, which is precisely why they are worth writing down. The walls documented below are structural properties of the venue and of transaction-cost arithmetic; they do not depend on our estimators, and anyone entering this market will hit them at full cost unless told where they are. Fifteen refutations at roughly ten thousand tested configurations is, in effect, a large prepaid experiment whose output is a map.

Three choices shape the paper:

1. **Organisation by mechanism, not by technique.** Hawkes intensities, order-flow imbalance, VPIN, jump detection, OU mean-reversion and book-pressure features are six different techniques — and all six draw on one economic source: short-horizon order-flow pressure in a single venue's book. Organised by technique, the record looks diversified; organised by *who pays*, the concentration is visible, and the failures stop being surprising. Two techniques applied to one edge source are not diversification.

2. **Death reasons are first-class.** "This failed" is not one statement but four (Section 5), and conflating them is how live questions get buried and dead ones exhumed. Each entry in the census carries its death reason, and only two of the four reasons close a question permanently.

3. **The measurements are the contribution; the graveyard is the service.** The descriptive facts in Section 3 — spread structure, depth, funding cadence, history retention, and the survivorship boundary of the data itself — hold independently of any trading hypothesis, are reproducible from the frozen slice shipped with this paper (Appendix A), and several of them contradict intuitions imported from equity microstructure or from centralised crypto venues.

One boundary is deliberate: this paper publishes methods, measurements and refutations. It does not publish anything the record currently treats as a live, unrefuted edge, and the census below should be read with that filter in mind.

2 The venue and the data

Hyperliquid is a fully on-chain perpetual-futures exchange with a central limit order book, permissionless API access over REST and WebSocket, an hourly funding clock (Section 3.3), and — unusually — publicly observable wallet positions and liquidation levels. At the time of the census the venue listed 177 tradeable perpetual pairs.

The internal platform ingests the venue's book and trade stream at 100 ms cadence into a 236-feature representation per symbol (BTC, ETH, SOL; ∼2.17 million ticks per symbol in the core sample), and maintains three slower archives across the full universe: OHLCV candles at 1m/5m/15m/1h (∼3.06 million candles, 708 series), periodic full-book snapshots for all 177 pairs, and, since August 2026, the hourly funding-rate history of every pair.

Survivorship, declared. The candle archive covers the 177 pairs listed during the collection window. At least 55 previously listed perpetuals were delisted before or during the window and are absent. Delisting concentrates in exactly the thin, low-attention tail where several of the untested mechanisms (notably gradual information diffusion, Section 6) predict their effect, so cross-sectional results below are conditioned on survival to an extent we cannot fully quantify. We state this rather than let a referee find it: every cross-sectional number in this paper should be read as "among pairs that survived to be measured."

History is not recoverable. The venue caps retrievable candle history at ∼5000 bars per interval regardless of start time. One-minute history therefore expires ∼3.5 days after the fact and *cannot be backfilled at any later date*; the same applies pro-rata to every interval. There is, to our knowledge, no public archive of this venue's L2 book states. Datasets here are accumulated live or lost, which is why the reproducibility slice (Appendix A) is frozen into the repository rather than re-fetched on demand.

3 Descriptive measurements

These facts constrain every strategy on the venue, are independent of any signal, and each one killed or reshaped at least one hypothesis in the census.

3.1 The universe is not its majors

Measured over repeated full-book snapshot sweeps of all 177 pairs, the cross-sectional median half-spread is 1.37–1.43 basis points against BTC's 0.077–0.083 bps: **the median pair quotes 17–19× wider than BTC**, and 169 of 177 pairs are wider than the three majors on which all high-frequency work was done. Any result established on BTC's book generalises to almost nowhere on the venue, and any cost model calibrated on the majors understates the universe by an order of magnitude.

3.2 Depth does not follow spread

Across the same sweeps, per-pair half-spread and touch depth are essentially uncorrelated in cross-section ($\rho = -0.107$): wide pairs are not deep pairs resting at a fairer price, they are simply thin. The median pair carries \$33–\$391 of notional at the best quotes (slice median \$81), and **only 4 of 177 pairs hold \$5,000 at the touch**. The extreme cases are instructive: pairs quoting 13–27 bps half-spreads over \$3–\$20 of touch notional. A large per-fill edge on \$20 of size is not a business. Admission to any cross-sectional strategy is therefore a *joint* spread–depth condition, and the joint condition is far more restrictive than either margin: a 2 bps half-spread ceiling alone admits 129 pairs, but the capacity to deploy even \$1,000 per pair per day at 1% of volume cuts the tradeable set to ∼117, and \$5,000 of touch size exists in four names.

3.3 Funding settles hourly, and the displayed rate is hourly

Perpetual funding on this venue is exchanged **every hour** (verified directly: settlement timestamps in the venue's own funding history are spaced 1.0000 h apart), and the funding field in the live asset context is the hourly rate — observed baseline 1.25×10^{-5} per hour, i.e. the venue's 0.00125%/h default. Conventional CEX intuition (8-hour funding) mis-prices held positions on this venue by a factor of eight. In our own simulation stack this exact error survived undetected for months — funding was charged in *no* backtest, and the configured interval was 8 hours — an error we document in Section 4 because the manner of finding it generalises.

3.4 Retention, cadence, and what they imply

The ∼5000-bar retention cap (Section 2) has a strategic consequence that is easy to miss: for any research programme on this venue, *data continuity is the binding constraint*, ahead of ideas, compute, or capital. A gap in live collection is permanent. Half of the operational machinery described in Section 4 exists to protect continuity, and the single most damaging defect we record (Section 6, entry M) was an availability failure, not a statistical one.

4 Method, and what the method caught

The statistical equipment is standard; what we would emphasise is the combination, because in every near-miss of the final month it was a *control*, not judgement, that killed the hypothesis.

- **Planted-signal tests before real data.** Every estimator must first recover a synthetic effect of known size planted in synthetic data, and return null on planted noise. Three separate estimator bugs — each of which would have manufactured a discovery — were caught only this way.
- **Null calibration under the right exchangeability.** Permutation nulls are constructed to preserve the dependence structure the statistic is exposed to (day-block permutations for serial dependence, label permutations that preserve group sizes for cross-sectional claims). One internal audit produced an IC of 0.39–0.46 — *higher* than the claim it was auditing — purely from overlapping forward windows; non-overlapping sampling is mandatory since.
- **False-discovery control across the whole grid.** Benjamini–Hochberg at $q = 0.05$ over every cell actually computed, not over the cells that looked interesting.
- **Per-day durability, not pooled means.** A pooled mean over 25 days is routinely carried by two days. Every claim must hold on a stated fraction of individual days.
- **Pre-registration in version control.** Acceptance criteria, trial counts for deflated-Sharpe penalties (in the sense of Bailey and López de Prado, 2014), and stop rules are committed to the repository *before* a study runs; the commit hash is the proof. The funding-carry study of entry O below is the worked example: its gate, criteria, and sceptical priors were committed before its data was fetched, and the study's own gate was not permitted to be adjusted afterwards.
- **Cost realism as a gate, not a sensitivity.** All costs flow from one configuration file through one loader. The census contains a five-fold false-discovery event (entry C) whose root cause was a silently wrong venue's fee tier reaching a harness default.

Two of the fifteen entries below are methodological deaths — the hypothesis was never really tested because the harness was wrong — and we count them in the census deliberately. A negative-results paper that omitted its own instrumentation failures would be reporting a cleaner record than the one that exists.

5 How a mechanism dies

Death reason	Permanent?	Implication
Arithmetic — predicted move ≈ cost	Yes, at that horizon	Only a cost change reopens it
Mechanism absent — measured, not there	Yes	Closed
Undecidable — underpowered	No	Revisit at adequate n; a *collection* task, not a research one
Blocked — data or access missing	No	Acquire the data; the question is untested, not answered

Table 1: The four death reasons. Only the first two close a question. Filing an underpowered result as "refuted" discards a live question; filing an arithmetic death as "needs more data" wastes a year. The census tags every entry with one of these four.

The distinction earns its keep twice in this record: two of the strongest-looking refutations (entries J and L) are formally *undecidable* — one observed 1.5 events per day against a design that needed hundreds, the other needs ∼0.9 years of rebalances for its Sharpe ratio to reach significance — and treating either as "mechanism absent" would have been a classification error with multi-month consequences in both directions.

6 The census

Table 2 is the record. Family numbers refer to the economic taxonomy of Section 7.

Table 2: The fifteen-entry census. "Family" refers to Table 3. Entries C and M are instrumentation deaths and are included deliberately (Section 4).

#	What was tried	Outcome	Family	Death reason
A	Taker capture of the 1–5 s imbalance signal	0.5–2 bps predicted move vs 11 bps round trip	1/11	Arithmetic
B	Naive maker capture of the same signal	IC 0.45 → 0.03 conditional on directionally-consistent fills	1	Arithmetic (adverse selection)
C	Five production signal "winners"	All five refuted on re-audit	—	Methodological (wrong venue's fee tier in harness defaults)
D	VWAP-reversion at taker cost, 58 days	−25k to −41k bps cumulative	1/11	Arithmetic (∼50 trades/day × fees)
E	Textbook Avellaneda–Stoikov quoting	−1.89 bps/fill; risk-widened quotes rest behind the touch and fill only through adverse moves	1	Model-transfer failure
F	Touch-pegged passive quoting, 8 configs × 179 symbol-days	0 survive day-consistency and concentration	1	Mechanism absent (at the majors' touch)
G	Fee-tier staking discounts as the fix	Verdict tier-invariant: discounts apply to fees, never rebates	1	Arithmetic

#	What was tried	Outcome	Family	Death reason
H	The full maker rebate ladder	Breakeven rebate +0.144 bps exceeds the best volume-tier rate (zero); first viable rung requires ∼1.5% of venue-wide maker volume	1	Arithmetic (structural)
I	Momentum persistence, 1m/5m bars, majors	Anti-persistent: 34/36 cells negative; the universe mean-reverts ∼8-to-1	9	Mechanism absent (naked signature, bar scale)
J	VWAP-band oscillation harvesting, 330 cells	0 survive — but ∼1.5 events/day against a design needing hundreds	1	*Undecidable*
K	Permutation entropy as cross-sectional score	Interquartile range 0.0005: the middle half of the universe is indistinguishable	—	Mechanism absent
L	Cross-sectional low-vol rotation (long/short, out of sample)	0/6 pre-registered criteria; beta-neutral rebuild reaches 4/6 with Sharpe 2.1 but needs ∼0.89 yr for significance	8/9	*Undecidable*
M	Hierarchical signal combiner	Loses to one of its own inputs; weights were fitted after the window they were scored on	—	Methodological
N	Agreement gating (trade only when signals agree)	Harmful: the *disagreement* subset carries the signal	2	Mechanism absent (as specified)
O	Funding carry, dollar-neutral across 118 pairs, 92 days, pre-registered	Funding leg collects +5–11% gross; the short-the-crowd price leg loses more; 0/6 configs pass	4	Mechanism absent (carry ≈ fair drift compensation)

Three entries deserve prose beyond the table.

B: the execution boundary. The strongest structure in the record — directional IC ≈ 0.45 at seconds — collapses to ≈0.03 when conditioned on obtaining a directionally-correct passive fill, while conditioning on the presence of *any* fill *raises* the coefficient to 0.52. The information is real and the market microstructure prices it: a passive order fills preferentially when the move is against it. This isolates adverse selection, not signal decay, as the binding constraint, and it is the single result that most reshaped the programme.

H: zero fees are not free money. At BTC's touch the half-spread (0.077–0.083 bps) is roughly one third of the expected adverse move conditional on a fill (0.23–0.24 bps), so a resting quote loses ∼0.08 bps per posting even at a zero fee rate. The venue must *pay* ∼ + 0.15 bps before passive quoting at the majors' touch breaks even, and the first rebate rung that clears this sits at ∼1.5% of venue-wide 14-day maker volume (∼$86M/day of maker fills at current venue volume) — an entry ticket, not a fee schedule. The reproduction script regenerates this ladder from the frozen slice and the venue's published tiers.

O: the carry that was exactly fair. The funding-carry study is the census's worked example of the full method: mechanism named in advance (the crowded side pays hourly; a dollar-neutral book collects the spread), gate and criteria committed to version control before the data existed, and a cost-ratio gate (expected carry over expected churn cost ≥ 3) evaluated *before* the backtest was permitted to run. The gate passed at 4.5; the book still failed. Funding was collected almost exactly as forecast (+5–11% gross over 92 days) and the price leg gave it back with interest: the coins whose crowd paid the most kept moving the crowd's way. On this venue, over this window, funding was a fair price for drift — a textbook risk-premium story, measured rather than assumed. The two marginally positive slow configurations are disqualified rather than suggestive: the best carries 92% of its P&L in one day and would need ∼4.6 years for its Sharpe ratio to clear significance.

7 The mechanism families, and what remains open

Table 3: Families by economic source. Numbers are stable identifiers in the internal taxonomy; 6, 10, 12 were re-specified or retired and never reused — gaps are deliberate. Two families are omitted as operationally inaccessible (volatility risk premium: no options market; queue-priority/latency: no colocation).

#	Family — economic source	Status after the census
1	Liquidity provision — paid to bear inventory + adverse selection (cf. Grossman and Miller, 1988; Avellaneda and Stoikov, 2008)	Heavily worked; closed at the majors' touch by arithmetic (entries E–H); open on wide pairs, where the half-spread is a multiple of BTC's
2	Adverse-selection avoidance (cf. Kyle, 1985; Easley et al., 2012)	Survives as a *gate* only: flow-toxicity conditioning lifts per-fill economics on all three majors but carries no direction
3	Price-discovery lag across venues (cf. Hasbrouck, 1995; Makarov and Schoar, 2020)	Untried — blocked on a cross-venue feed
4	Funding reflexivity	Carry branch measured and dead (entry O); the reflexive-unwinding branch (funding → forced flow → price) untested, blocked on position-history accrual
5	Deterministic liquidation — the engine executes at a computable price (nearest analogue Coval and Stafford, 2007)	Best-supported untried family: trigger prices are *public* on this venue; blocked on instrumentation, not theory
6	Gradual information diffusion (cf. Hong and Stein, 1999; Hong et al., 2000)	Untried at its native horizon (weeks); the minute-scale reversion result does not touch it, and the pooled tests that exist would dilute a liquidity-conditional effect to zero by construction
7	Attention and flow-driven demand (cf. Barber and Odean, 2008)	Untried; listing events exist in-archive but $n = 2$ observable events so far — a collection task
8	Statistical relative value (cf. Gatev et al., 2006)	Registered, never evaluated
9	Behavioural under/over-reaction (cf. Jegadeesh and Titman, 1993; De Bondt and Thaler, 1985)	The *naked signature* is exhausted at bar scale (entry I); the generating mechanisms are largely untried
11	Microstructure noise / bid–ask bounce (cf. Roll, 1984)	Exhausted

Two remarks the table cannot carry.

Signatures are not mechanisms. Momentum and mean-reversion are signs of autocorrelation — observables, not explanations. At least four distinct mechanisms produce reversion (inventory compensation, forced-flow overshoot, bid-ask bounce, behavioural over-reaction) and four produce momentum (discovery lag, gradual diffusion, attention flow, under-reaction). What this record exhausted is the *naked signature* — trade the sign of recent returns, at bar scale, at taker cost — which says nothing about the generating mechanisms individually. The question to ask any momentum or reversion proposal is not "what horizon?" but "*paid by whom?*"

The strongest negative is a signpost. "The universe reverts 8-to-1" (entry I) is precisely the condition under which liquidity provision and forced-flow harvesting pay, because both are compensated *by* reversion. The most robust refutation in the record therefore points directly at the two families the venue's architecture makes most observable — and both are blocked on instrumentation rather than closed. On-chain position visibility is a structural research advantage of this venue class: forced flow that equity-market studies must infer from returns is, here, directly observable in advance.

8 The transferable result

Every arithmetic death in Table 2 was knowable in advance. The predicted move of the core signal (0.5–2 bps at seconds) against the realistic round trip (∼11 bps taker) is a ratio of ∼0.1; the programme's subsequently adopted rule — *expected move over round-trip cost* ≥ 3 *at the design horizon, computed before anything is built* — would have failed the entire high-frequency taker programme on day one, using twenty minutes and public data. The same twenty minutes, run at gate zero of the funding-carry study (entry O), correctly admitted it — and the study then failed for a reason the ratio cannot see, which is the right division of labour: the ratio filters what cannot work; measurement decides what does not.

Stated for the general case: on any venue, for any signal, compute m/c where m is the expected favourable move over the holding horizon and c the full round trip at the fee tier you actually hold. Below 1, the strategy is arithmetically dead regardless of the information content; between 1 and 3, it is a fill-quality research programme, not a trading strategy; above 3, it earns the right to a backtest. The census contains six entries that this one-line calculation would have prevented.

Acknowledgements

The platform, studies and record summarised here were built and run by the author. No external funding; no conflicts to declare.

A Reproducibility

The repository ships a frozen public slice and a one-command reproduction path:

```
./reproduce.sh
```

requiring only Python with `pandas`, `pyarrow` and `matplotlib`; no network access and no credentials. The script reads the frozen slice (two full-universe L2 snapshot days — 177 pairs, ∼50k book states, ∼9.9 MB — plus hourly candles for the majors) and the venue fee configuration, and regenerates:

1. the cross-sectional half-spread distribution (Section 3.1): slice values — median 1.428 bps, 18.6× BTC;
2. the touch-depth distribution (Section 3.2): slice values — median \$81; 4/177 pairs at \$5,000;

3. the maker-ladder breakeven (entry H): slice values — breakeven rebate +0.151 bps; first viable rung unchanged.

together with a machine-readable `headlines.json`. The slice's own values are pinned by unit tests, so any drift in the computation is a build failure rather than a silent revision. Two constants (expected adverse selection conditional on a fill at BTC's touch, 0.228/0.242 bps bid/ask) enter the ladder figure from tick data too large to freeze and are declared as frozen inputs in the script's header. Because the venue caps history retention and serves no historical order-book states (Section 2), freezing the slice in the repository is the only mechanism by which these measurements remain independently checkable.

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